

IPv4 (INTERNET PROTOCOL VERSION 4)

WHAT IS IPv4 ?

IPv4 (Internet Protocol Version 4) is a Layer 3 protocol used to uniquely identify devices on a network and provide communication between different networks.

Without IPv4, devices would not know:

- Who is sending the data.
- Who should receive the data.
- Which path the data should take.

Example:

PC1 → 192.168.1.10

PC2 → 192.168.1.20

When PC1 sends data, it places:

Source IP : 192.168.1.10

Destination IP : 192.168.1.20

The router uses these addresses to forward packets.

WHY DO WE NEED IPv4?

Imagine the Internet without addresses.

Suppose there are billions of devices:

- Computers
- Laptops
- Mobile Phones
- Servers
- Printers
- Cameras

If every device had no address:

- Data would get lost.
- Devices couldn't find each other.
- Internet communication would be impossible.

Therefore every device needs a unique logical address.

This logical address is called an IP Address.

LOGICAL ADDRESS VS PHYSICAL ADDRESS

Physical Address = MAC Address

Logical Address = IP Address

Example:

MAC Address:

00:1A:2B:3C:4D:5E

IP Address:

192.168.1.10

MAC identifies a device inside a LAN.

IP identifies a device across multiple networks.

IPV4 ADDRESS STRUCTURE

IPv4 Address Size = 32 Bits

Example:

192.168.1.10

Each section is called an Octet.

192 . 168 . 1 . 10

8	8	8	8

Total:

$8 + 8 + 8 + 8 = 32$ Bits

IPV4 ADDRESS CLASSES

Class	1st octet of IP address	Default Subnet Mask	Network / Host	Number of networks	Maximum nodes in a network
A	1 - 126	255.0.0.0	N.H.H.H	126	16,777,214
B	128 - 191	255.255.0.0	N.N.H.H	16,384	65,534
C	192 - 223	255.255.255.0	N.N.N.H	2,097,152	254
D	224 - 239				
E	240 - 254				

IPV4 ADDRESS PARTS

Every IPv4 Address contains:

1. Network Portion
2. Host Portion

Example:

192.168.1.10/24

Network Portion:

192.168.1

Host Portion:

10

Network Portion identifies the network.

Host Portion identifies the device.

WHAT IS A SUBNET MASK?

A Subnet Mask is a 32-bit number used to identify which part of an IP address belongs to the Network and which part belongs to the Host.

Think of it as a separator.

IP Address tells us:

“Who am I?”

Subnet Mask tells us:

“Which network do I belong to?”

Example:

IP Address:

192.168.1.10

Subnet Mask:

255.255.255.0

Result:

Network Portion = 192.168.1

Host Portion = 10

Without a subnet mask, a device cannot determine where the network portion ends and the host portion begins.

WHAT IS CIDR NOTATION?

CIDR stands for:

Classless Inter-Domain Routing

CIDR is simply a shorter way to write subnet masks.

Example:

255.0.0.0 = /8

255.255.0.0 = /16

255.255.255.0 = /24

Instead of writing the full subnet mask every time, engineers usually write CIDR notation.

Example:

192.168.1.10/24

This is easier and faster to read.

IPV4 ADDRESS TYPES

A. Public IP Address

Used on Internet.

Example:

8.8.8.8

Public IPs are globally unique.

B. Private IP Address

Used inside organizations.

Ranges:

Class A:

10.0.0.0 – 10.255.255.255

Class B:

172.16.0.0 – 172.31.255.255

Class C:

192.168.0.0 – 192.168.255.255

Private IPs cannot be routed directly on Internet.

SPECIAL IPV4 ADDRESSES

Network Address

Example:

192.168.1.0/24

Represents entire network .

Cannot assign to host.

Broadcast Address

Example:

192.168.1.255/24

Used to communicate with all devices.

Cannot assign to host.

Loopback Address

127.0.0.1

Used for testing TCP/IP stack.

Command:

ping 127.0.0.1

APIPA (Automatic Private IP Addressing)

169.254.0.0/16

Assigned automatically when DHCP fails.

Example:

169.254.10.20

DEFAULT GATEWAY

A Default Gateway is the router used to reach other networks.

Example:

PC IP:

192.168.1.10

Gateway:

192.168.1.1

Destination:

10.10.10.10

Since destination is outside local network:

PC sends packet to Gateway.

Gateway forwards packet.

HOW IPV4 WORKS

Step 1:

User opens Google.

Step 2:

DNS converts:

google.com

to

142.250.x.x

Step 3:

PC creates packet.

Source IP:

192.168.1.10

Destination IP:

142.250.x.x

Step 4:

Packet sent to Router.

Step 5:

Routers forward packet through Internet.

Step 6:

Google server receives packet.

Step 7:

Response comes back.

This entire communication depends on IPv4.

PROBLEM WITH CLASSES

Suppose a company needs:

500 Hosts

Class C provides:

254 Hosts

Not enough

Class B provides:

65,534 Hosts

Too many

Thousands of IP addresses are wasted.

This problem led to the development of Subnetting and CIDR.

WHAT IS SUBNETTING ?

Subnetting is the process of dividing one large network into multiple smaller networks called Subnets.

Subnet = Sub + Network

Subnet = Smaller Network

REAL-WORLD EXAMPLE

Imagine a company building.

Building = Network

Departments = Subnets

Employees = Hosts

Without departments:

Everyone works in one large hall.

With departments:

People are organized and communication becomes easier.

Subnetting does the same thing in networking.

HOW SUBNETTING WORKS

Subnetting works by borrowing bits from the Host Portion and converting them into Network Bits.

Example:

Original Network:

192.168.1.0/24

Network Bits = 24

Host Bits = 8

After Subnetting:

192.168.1.0/26

Network Bits = 26

Host Bits = 6

Two Host Bits were borrowed and converted into Network Bits.

WHY IS SUBNETTING REQUIRED?

Subnetting is required because large networks create problems.

Problems of a Large Network:

- More Broadcast Traffic
- Slow Network Performance
- Difficult Troubleshooting
- Difficult Management
- Poor Security

Subnetting solves these problems by creating smaller networks.

BENEFITS OF SUBNETTING

- Reduces Broadcast Traffic
 - Improves Network Performance
 - Improves Security
 - Better Network Organization
 - Easier Troubleshooting
 - Efficient IP Address Utilization
 - Easier Management
 - Supports Network Growth
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NETWORK ADDRESS

The first address of every subnet is called the Network Address.

Example:

Network:

192.168.1.0/24

Network Address:

192.168.1.0

Purpose:

Identifies the subnet itself.

Cannot be assigned to a host.

BROADCAST ADDRESS

The last address of every subnet is called the Broadcast Address.

Example:

192.168.1.255

Purpose:

Used to send traffic to all hosts in the subnet.

Cannot be assigned to a host.

USABLE HOST RANGE

Usable Host Range means all addresses that can be assigned to devices.

Example:

Network Address:

192.168.1.0

Broadcast Address:

192.168.1.255

Usable Range:

192.168.1.1

to

192.168.1.254

MAGIC NUMBER

Magic Number is the increment between subnet networks.

Formula:

Magic Number = 256 - Interesting Octet

The Interesting Octet is the octet where the subnet mask changes from 255.

SUBNETTING CALCULATIONS

When a network engineer receives an IP address and subnet mask, he should be able to determine:

- Network Address = First Address
- Broadcast Address = Next Network - 1
- First Usable Host = Network + 1
- Last Usable Host = Broadcast - 1
- Total Usable Hosts = $2^{(\text{Host Bits})} - 2$
- Total Subnets = $2^{(\text{Borrowed Bits})}$
- Magic Number = $256 - \text{Interesting Octet}$

These are the most important subnetting calculations.

WHAT IS FLSM ?

FLSM stands for:

Fixed Length Subnet Mask

In FLFSM, every subnet uses the same subnet mask.

All subnets have:

- Same subnet size
- Same number of hosts
- Same subnet mask

EXAMPLE OF FLFSM

Suppose we have:

192.168.1.0/24

And we divide it into:

192.168.1.0/26

192.168.1.64/26

192.168.1.128/26

192.168.1.192/26

All subnets use:

/26

All subnets have:

62 Usable Hosts

This is called FLSM.

PROBLEM WITH FLSM

Imagine:

HR Department = 10 Hosts

Finance Department = 20 Hosts

Sales Department = 50 Hosts

IT Department = 60 Hosts

Using FLSM:

Every subnet gets 62 Hosts.

Result:

Many IP addresses are wasted.

WHAT IS VLSM ?

VLSM stands for:

Variable Length Subnet Mask

In VLSM, different subnet masks are used for different networks according to requirements.

Some subnets can be large.

Some subnets can be small.

WHY VLSM IS USED ?

To avoid IP address wastage.

To allocate IP addresses efficiently.

To support real-world network design.

VLSM EXAMPLE

Available Network:

192.168.1.0/24

Requirements:

IT Department = 100 Hosts

Sales Department = 50 Hosts

Finance Department = 25 Hosts

HR Department = 10 Hosts

STEP 1

Arrange requirements from largest to smallest.

100 Hosts

50 Hosts

25 Hosts

10 Hosts

STEP 2

Assign suitable subnet masks.

100 Hosts → /25

50 Hosts → /26

25 Hosts → /27

10 Hosts → /28

STEP 3

Allocate networks.

IT:

192.168.1.0/25

Hosts:

126

Sales:

192.168.1.128/26

Hosts:

62

Finance:

192.168.1.192/27

Hosts:

30

HR:

192.168.1.224/28

Hosts:

14

RESULT

No major IP wastage.

Network becomes efficient.

This is VLSM.

FLSM VS VLSM

FLSM

- Same subnet mask everywhere
- Easy to configure
- Wastes IP addresses
- Less efficient

VLSM

- Different subnet masks
 - More efficient
 - Saves IP addresses
 - Used in modern networks
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